MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, CROSSLINKED MODIFIED POLYETHYLENE (XLPE) INSULATED, 125°C, 600 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.

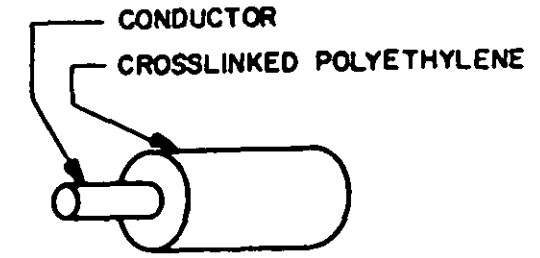


FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material	Coating		Min	Max
32	1 X 32	Copper	Tin	0.0080	0.024	0.030
32	7 X 40	Copper	Tin	.010	.026	.032
30	1 X 30	Copper	Tin	.0100	.026	.032
30	7 X 38	Copper	Tin	.012	.028	.034
28	1 X 28	Copper	Tin	.0126	.029	.035
28	7 X 36	Copper	Tin	.015	.031	.037
26	1 X 26	Copper	Tin	.0159	.032	.038
26	7 X 34	Copper	Tin	.019	.035	.041
26	19 X 38	Copper	Tin	.020	.035	.041
26	19 X 38	Copper	Silver	.020	.035	.041

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details. Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material	Coating		Min	Max
24	1 X 24	Copper	Tin	0.0201	0.036	0.044
24	7 X 32	Copper	Tin	.024	.040	.047
24	19 X 36	Copper	Tin	.025	.040	.047
22	1 X 22	Copper	Tin	.0254	.041	.049
22	7 X 30	Copper	Tin	.030	.046	.053
22	19 X 34	Copper	Tin	.032	.046	.053
20	1 X 20	Copper	Tin	.0320	.048	.055
20	7 X 28	Copper	Tin	.038	.054	.061
20	19 X 32	Copper	Tin	.040	.054	.061
18	1 X 18	Copper	Tin	.0403	.056	.064
18	7 X 26	Copper	Tin	.048	.064	.071
18	19 X 30	Copper	Tin	.050	.064	.071
16	1 X 16	Copper	Tin	.0508	.067	.075
16	19 X 29	Copper	Tin	.057	.073	.081
14	1 X 14	Copper	Tin	.0641	.080	.088
14	19 X 27	Copper	Tin	.072	.088	.096

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 3.4 kilovolts (kV).
Impulse dielectric test: 8.0 kV.
Tank test: 2.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 10,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 55°C ± 1°C (see table II).

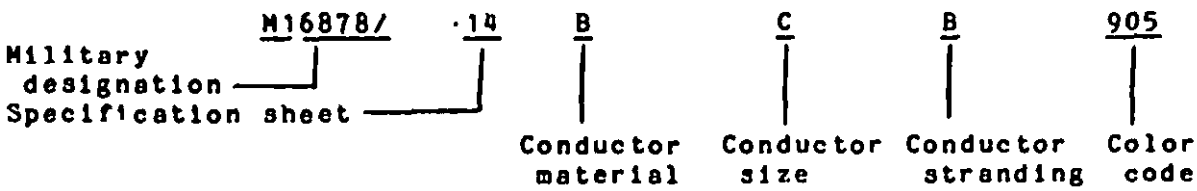
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
32 through 26	1
24 through 14	2

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 150°C.
Flammability: Applicable - 30 seconds (maximum) burning time
3 inches (maximum) flame travel
Soldering: Applicable.
Heat aging: 25 percent change (maximum) in elongation 96 hours at 135°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):



Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-13)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.